STAT 5703 **Name & UNI: Key**

Spring 2025

Midterm

Instructions:

1. There are 5 questions on this test.
2. Show all work.
3. Approved: One-page one-sided formula sheet, any calculator (without WiFi).
4. Tables: Attached.
5. If α is not provided, then the hypothesis test must be done at the 5% level.
6. Do not stay too long at a part where you experience a lot of difficulties.

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1. True or False:
   a) In linear regression analysis, the estimators of the coefficients are the same regardless of if they are obtained by the least squares approach or the maximum likelihood method under the normal error distribution assumption. T

   b) When using the method of moments, if we must estimate *m* parameters, then we need to compute *m* moments. T

   c) Let $X_1, \ldots, X_n$ be an iid sample from a population with finite mean μ and finite variance σ^2. Then the variance of the sample mean is $\boldsymbol{Var(\overline{X}) = \frac{\sigma}{\sqrt{n}}}$ F

   d) The mean squared error of an estimator $\boldsymbol{\widehat{\theta}_n}$ is defined as $\boldsymbol{MSE = E\left\{\left[\widehat{\theta}_n - E(\widehat{\theta}_n)\right]^2\right\}}$ F

   e) If, on the same sample, the confidence level is decreased, the width of the CI will increase. F

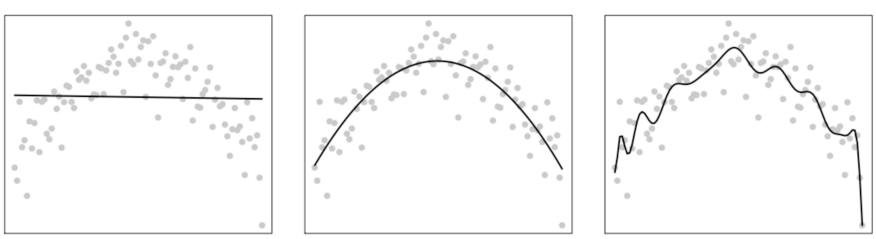
   f) The likelihood ratio test statistic Λ is always less than or equal to 1. T

   g) If we perform a hypothesis test with significance level 0.06 and compute the p-value for the same test, then the p-value would be less than 0.06. F

   h) The probability of Type II error (for any value of θ) is called the level of significance. F

   i) The Wald test statistic method is derived assuming the central limit theorem conditions are applicable to the sample used for the hypothesis test. T

   j) The graph below shows three regression models fitted on the same training data represented with gray dots. Model (3) has the smallest training error among them. T



$\widehat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$ (1) $\quad$ $\widehat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2$ (2) $\quad$ $\widehat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x + \cdots + \hat{\beta}_{20} x^{20}$ (3)

2. Let $X_1, \dots, X_n$ be a random sample from the pdf:

$$f(x;\theta) = (\theta+1)x^{\theta}, x \in [0,1]$$

where θ > -1 is an unknown parameter.

a) Find the method of moments estimator of θ.

$$E(X) = \int_0^1 x(\theta+1)x^{\theta}dx = (\theta+1)\int_0^1 x^{\theta+1}dx$$

$$= (\theta+1)\left.\frac{x^{\theta+2}}{\theta+2}\right]_0^1 = \frac{\theta+1}{\theta+2}$$

$$\frac{\theta+1}{\theta+2} = \bar{X} \Leftrightarrow \theta+1 = \theta\bar{X} + 2\bar{X} \Leftrightarrow \theta(1-\bar{X}) = 2\bar{X} - 1$$

$$\Rightarrow \hat{\theta}_{MM} = \frac{2\bar{X}-1}{1-\bar{X}}$$

b) Find the MLE estimator of θ.

$$L(\theta) = \prod_{i=1}^{n}\left[(\theta+1)X_i^{\theta}\right] = (\theta+1)^n\left(\prod_{i=1}^{n}X_i\right)^{\theta}$$

$$\Rightarrow l(\theta) = n\log(\theta+1) + \theta\log\prod_{i=1}^{n}X_i$$

$$\Rightarrow s(\theta) = l'(\theta) = \frac{n}{\theta+1} + \log\prod_{i=1}^{n}X_i = 0$$

$$\Leftrightarrow -\frac{n}{\log\prod_{i=1}^{n}X_i} = \theta+1$$

$$\Rightarrow \hat{\theta}_{MLE} = -\frac{n}{\log\prod_{i=1}^{n}X_i} - 1$$

c) Find the Fisher information $I_n(\theta)$.

$$l'(\theta) = \frac{n}{\theta + 1} + \log \prod_{i=1}^{n} X_i$$

$$\Rightarrow \; l''(\theta) = -\frac{n}{(\theta + 1)^2}$$

$$\Rightarrow I_n(\theta) = -E\big(l''(\theta)\big) = \frac{n}{(\theta + 1)^2}$$

d) Derive and simplify as much as possible the rejection region of the LRT for testing

H_0: $\theta = 0$ vs. H_a: $\theta \neq 0$

$$L(0) = (0 + 1)^n \left(\prod_{i=1}^{n} X_i\right)^0 = 1$$

$$L(\hat{\theta}_{MLE}) = \left(-\frac{n}{\log \prod_{i=1}^{n} X_i} - 1 + 1\right)^n \left(\prod_{i=1}^{n} X_i\right)^{-\frac{n}{\log \prod_{i=1}^{n} X_i} - 1}$$

$$= \frac{\left(-\frac{n}{\log \prod_{i=1}^{n} X_i}\right)^n \left(e^{\log \prod_{i=1}^{n} X_i}\right)^{-\frac{n}{\log \prod_{i=1}^{n} X_i}}}{\prod_{i=1}^{n} X_i}$$

$$= \frac{\left(-\frac{n}{\log \prod_{i=1}^{n} X_i}\right)^n e^{-n}}{\prod_{i=1}^{n} X_i}$$

$$\Rightarrow \Lambda = \frac{1}{\frac{\left(-\frac{n}{\log \prod_{i=1}^{n} X_i}\right)^n e^{-n}}{\prod_{i=1}^{n} X_i}} = \frac{e^n \prod_{i=1}^{n} X_i}{n^n \left(-\frac{1}{\log \prod_{i=1}^{n} X_i}\right)^n} \leq k$$

$$\Leftrightarrow \left(-\log \prod_{i=1}^{n} X_i\right)^n \prod_{i=1}^{n} X_i \leq k_1$$

3. Let $X_1, \dots, X_n$ be a random sample from a population $X \sim N(0, \sigma)$ with pdf $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}}$.

You may use without proof the facts that the MLE of σ^2 is $\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^n X_i^2$ and $E(X^{2k}) = \frac{(2k)!\sigma^{2k}}{k!2^k}$

a. Prove that

$$\frac{\sqrt{n}(\hat{\sigma}^2_{\text{MLE}} - \sigma^2)}{\sqrt{2\sigma^4}} \xrightarrow{d} N(0,1)$$

By CLT applied to the random variables $X_1^2, \dots, X_n^2$, we have for their average that:

$$\frac{\frac{1}{n}\sum_{i=1}^n X_i^2 - E(X^2)}{\sqrt{\frac{Var(X^2)}{n}}} \xrightarrow{d} N(0,1) \Leftrightarrow \frac{\sqrt{n}(\hat{\sigma}^2_{\text{MLE}} - \sigma^2)}{\sqrt{E(X^4) - [E(X^2)]^2}} \xrightarrow{d} N(0,1)$$

$$\Leftrightarrow \frac{\sqrt{n}(\hat{\sigma}^2_{\text{MLE}} - \sigma^2)}{\sqrt{\frac{(4)!\,\sigma^4}{2!\,2^2} - \sigma^4}} \xrightarrow{d} N(0,1) \Leftrightarrow \frac{\sqrt{n}(\hat{\sigma}^2_{\text{MLE}} - \sigma^2)}{\sqrt{2\sigma^4}} \xrightarrow{d} N(0,1)$$

b. Derive the form of the rejection region for the most powerful test of

$$H_0: \sigma^2 = 1 \; vs. \; H_a: \sigma^2 = 2$$

$$L(\sigma^2) = \prod_{i=1}^n \left[\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x_i^2}{2\sigma^2}}\right] = (2\pi)^{-\frac{n}{2}}(\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^n x_i^2}$$

$$\Rightarrow \frac{L(1)}{L(2)} = \frac{(2\pi)^{-\frac{n}{2}}(1)^{-\frac{n}{2}} e^{-\frac{1}{2(1)}\sum_{i=1}^n x_i^2}}{(2\pi)^{-\frac{n}{2}}(2)^{-\frac{n}{2}} e^{-\frac{1}{2(2)}\sum_{i=1}^n x_i^2}} = 2^{\frac{n}{2}} e^{-\frac{1}{4}\sum_{i=1}^n x_i^2} \le k$$

$$\Leftrightarrow e^{-\frac{1}{4}\sum_{i=1}^n x_i^2} \le k_1 \Leftrightarrow -\frac{1}{4}\sum_{i=1}^n X_i^2 \le k_2$$

$$\Leftrightarrow \sum_{i=1}^n X_i^2 \ge c$$

c. Regardless of your answer to part c), suppose n = 30 and we want to use the rejection region $RR = \{\sum_{i=1}^n X_i^2 \ge c\}$ with α = 0.05.

i) Find c exactly using the chi-square table.

$$P(\textstyle\sum_{i=1}^n X_i^2 \ge c | H_0) = 0.05 \Leftrightarrow P(\chi^2_{30} \ge c) = 0.05 \Leftrightarrow c = 43.773$$

ii) Find c approximately using the asymptotic result from part b) and the z-table.

$$P(\textstyle\sum_{i=1}^n X_i^2 \ge c | H_0) = P\left(\frac{\sqrt{n}(\hat{\sigma}^2_{\text{MLE}} - \sigma^2)}{\sqrt{2\sigma^4}} \ge \frac{\sqrt{30}\left(\frac{c}{30} - 1\right)}{\sqrt{2}}\right) \approx P\left(Z \ge \frac{\sqrt{30}\left(\frac{c}{30} - 1\right)}{\sqrt{2}}\right)$$

$$= 0.05 \Leftrightarrow \frac{\sqrt{30}\left(\frac{c}{30} - 1\right)}{\sqrt{2}} = 1.645 \Leftrightarrow c = 30\left(\frac{1.645(\sqrt{2})}{\sqrt{30}} + 1\right) = 42.742$$

4. In a recent poll 50 randomly selected people were asked "Do you own a smart phone?". Of these respondents 49 replied "Yes" and only 1 replied "No". Suppose we want to estimate with a 75% confidence interval the true proportion *p* of all people who own a smart phone.

a) Construct the interval with the classic formula $\hat{p} \pm z_{\frac{\alpha}{2}}\widehat{SE}(\hat{p})$, where $\hat{p} = \frac{x}{n}$.
Note: You may use directly, without deriving it, the formula for $\widehat{SE}(\hat{p})$ from class.

$$\hat{p} = \frac{49}{50} = 0.98, \widehat{SE}(\hat{p}) = \sqrt{\frac{0.98(0.02)}{50}} = 0.0198$$

$Z_{0.125} = 1.15$
75% CI for *p* is $.98 \pm 1.15(0.0198) = 0.98 \pm 0.02277 = (0.957, 1.003)$

b) Construct the interval with the formula $\tilde{p} \pm z_{\frac{\alpha}{2}}\widehat{SE}(\tilde{p})$, where $\tilde{p} = \frac{x+1}{n+2}$.
Note: You must derive the formula for $\widehat{SE}(\tilde{p})$ even if you know/have it.

$$\hat{p} = \frac{49+1}{50+2} = \frac{50}{52} = 0.9615$$
$$Var(\tilde{p}) = Var\left(\frac{X+1}{n+2}\right) = \frac{Var(X)}{(n+2)^2} = \frac{np(1-p)}{(n+2)^2}$$

$$\widehat{SE}(\tilde{p}) = \sqrt{\frac{50(.9615)(1-0.9615)}{(52)^2}} = 0.026$$

Note: Some students might estimate $\widehat{SE}(\tilde{p}) = \sqrt{\frac{50(.98)(1-0.98)}{(52)^2}} = 0.019$ which is somewhat reasonable and should be accepted.

75% CI for *p* is $.9615 \pm 1.15(0.026) = .9615 \pm 0.03 = (0.9315, 0.9915)$

c) *Bonus*: For the classic estimator $\hat{p} = \frac{x}{n}$ there exists a result known as the Hoeffding's inequality, which states that

$$P(|\hat{p} - p| > c) \leq 2e^{-2nc^2}$$

Use the above result to construct and compute a 75% CI for *p*.

$$P(\hat{p} - c < p < \hat{p} + c) \geq 2e^{-2nc^2} = 0.75$$
$$\Rightarrow e^{-2nc^2} = 0.375 \iff 2(50)c^2 = \log 0.375 \iff c^2 = 0.0098 \Rightarrow c = 0.099$$

75% CI for *p* is $(0.98 - 0.099, 0.98 + 0.099) = (0.881, 1.079)$

5. Consider again the data from class about rowing times. There were 90 college athletes competing in a 2000 m event. For each athlete the following measurements were recorded: race time (in seconds), arm span (in cm), height (in cm), and an indicator preexper = 1 if the athlete had previous experience or 0 if they didn't have. Several linear regression models were fit to predict the race time from the other variables. The resulting R code is below:

```
> m1 = lm(racetime ~ armspan, data = rowtime)
> summary(m1)
lm(formula = racetime ~ armspan, data = rowtime)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 571.0368    45.2133  12.630   <2e-16 ***
armspan      -0.6510     0.2612  -2.492   0.0146 *
---
Residual standard error: 23.71 on 88 degrees of freedom
Multiple R-squared:  0.06593,                Adjusted R-squared:  0.05531
F-statistic: 6.211 on 1 and 88 DF,  p-value: 0.01457

> m2 = lm(racetime ~ armspan + tall, data = rowtime)
> summary(m2)
lm(formula = racetime ~ armspan + tall, data = rowtime)
 Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 835.4860    69.8273  11.965  < 2e-16 ***
armspan       0.1879     0.2960   0.635    0.527
height       -2.3683     0.5083  -4.660 1.13e-05 ***
---
Residual standard error: 21.33 on 87 degrees of freedom
Multiple R-squared:  0.2525,                  Adjusted R-squared:  0.2353
F-statistic: 14.69 on 2 and 87 DF,  p-value: 3.182e-06

> m3 = lm(racetime ~ tall + preexper, data = rowtime)
> summary(m3)
lm(formula = racetime ~ tall + preexper, data = rowtime)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 836.9171    65.2517  12.826  < 2e-16 ***
height       -2.1577     0.3772  -5.720 1.48e-07 ***
preexper1   -16.0803     4.4600  -3.605 0.000519 ***
---
Residual standard error: 19.94 on 87 degrees of freedom
Multiple R-squared:  0.3466,                  Adjusted R-squared:  0.3316
F-statistic: 23.08 on 2 and 87 DF,  p-value: 9.103e-09
```

a) Looking at models 1 and 2, choose all correct statements:

i. According to model 1, athletes with larger arm span were faster on average.
ii. According to model 2, athletes with larger arm span were faster on average.
iii. The direct causal effect of arm span is a significant predictor at α = 0.05 level.
iv. The total effect of arm span is a significant predictor at α = 0.05 level.

b) Give the most likely explanation of the discrepancies between models 1 and 2.

i. There is a programming error.
ii. There is strong evidence that the model assumptions are violated.
iii. These models overfit the data.
iv. There is a fork or pipe confounding effect between height, arm span and race time.

c) Choose all correct statements about model 3:

i. Everything else the same, athletes with previous experience were slower on average.
ii. Everything else the same, athletes with previous experience were faster on average.
iii. This is known as a model with interaction effect.
iv. This is a model that fits two parallel lines.

d) Write down the two fitted equations based on model 3.

$$\widehat{time} = 836.9 - 2.2 \times height - 16.1 \times preexp$$

With previous experience: $\widehat{time} = 820.8 - 2.2 \times height$

Without previous experience: $\widehat{time} = 836.9 - 2.2 \times height$

e) Choose the model with the smallest SSE and compute its value.

$$Model\ 3: SSE = (n - p - 1) \times MSE = 87(19.94\ )^2 = 34591.51$$

f) In model 3, estimate with a 90% confidence interval the change in the average time when the height increases by one cm, but everything else remains the same. Use your answer to decide if height is a significant predictor in the model. Use nearest df from the table.

df = 87, $t_{0.05} \approx 1.664$

90% CI for β_1 is $-2.16 \pm 1.664(0.38)$ = (-2.79, -1.53)

Since the CI does not contain 0, we reject H_0: $\beta_1 = 0$ and conclude that height is a significant predictor.

g) Circle all correct statements about model 3:

i. There are nonsignificant predictors at the 0.05 level.
ii. The adjusted R^2 will increase if we fit splines or polynomial terms instead of linear.
iii. The negative signs of the estimated coefficients make no common sense.
iv. The race time errors are estimated to have a standard deviation of less than a minute.

Standard Normal Probabilities

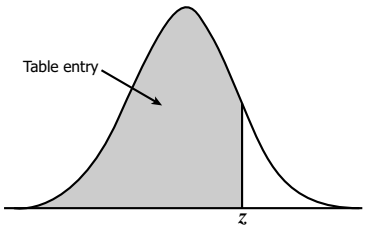


Table entry for z is the area under the standard normal curve to the left of z.

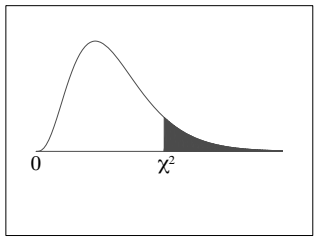
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

t-table.xls 7/14/2007

Chi-Square Distribution Table



The shaded area is equal to α for $\chi^2 = \chi^2_\alpha$.

df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169